TOMATO FERTILITY REPORT ANALYSIS

TOK2322C_Tomato_Fertility_Report.xlsx

Report ID: TOK2322 | **Protocol:** BOW11-22-33
Trial Type: Agricultural Field Trial | **Crop:** Tomato
Report Date: December 6, 2023 | **Organization:** Pacific Ag Research

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EXECUTIVE SUMMARY

This is a **HYBRID REPORT** with approximately **70% AUTOMATIC and 30% MANUAL** components:

- **Automatic Generation:** Data processing, formula calculations, statistical tables, charts
- **Manual Work:** Data collection, analysis interpretation, narrative writing, quality checks

The report uses a **systematic template-based system** with built-in automation, but requires significant manual field data collection and expert interpretation.

REPORT STRUCTURE & CONTENT OVERVIEW

📊 **Total Sheets: 10**

Sheet Name	Type	Purpose	Automation Level
Narrative	Summary	Key findings & conclusions	0% (Manual)
Trial Information	Reference	Metadata & trial parameters	Manual entry
Treatment List & Map	Reference	Treatment details & field layout	50% (Hybrid)
Assessment Data Summary	Raw Data	Original field measurements	10% (Data input)
Chartwork	Calculation	Derived data with formulas	80% (Automatic)
Figures	Visualization	Chart embeddings	90% (Auto-generated)
Chartwork	Visualization	Statistical charts	90% (Auto-generated)

Sheet Name	Type	Purpose	Automation Level
Post-Harvest Chartwork	Analysis	Storage/quality data graphs	90% (Auto-generated)
AOV Means Table	Statistics	Analysis of Variance results	95% (Automatic)
Weather	Reference	Environmental conditions	Manual data import
Photos	Documentation	Trial photographs	Manual

DETAILED ANALYSIS BY SHEET

1. NARRATIVE SHEET (0% Automatic - Pure Manual)

Purpose: Executive summary and key findings

Content:

- Key Findings paragraph (test results summary)
- Summary section (detailed narrative of trial conditions and methods)
- Conclusions and observations

Process:

- ✋ **100% Manual** - Written by study director/investigator
- Requires expert interpretation of all data
- Evidence-based conclusions from analysis
- Professional agricultural science writing

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2. TRIAL INFORMATION SHEET (20% Automatic)

Purpose: Complete trial metadata and setup information

Content:

```
Trial ID: TOK2322
Protocol ID: BOW11-22-33
Client Contact: Rhoam Bosphoramus
Sponsor: Hyrule Scientific
# Treatments: 2
# Replicates: 6
Design: RCB (Randomized Complete Block)
Dates: June 30 - September 16, 2023
```

Process:

- ✋ **80% Manual** - Input during trial initiation
- 20% Automated - Drop-down selections from database
- Locked fields after initial setup
- Merged cells (12) for formatting

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3. TREATMENT LIST & MAP (50% Automatic)

Purpose: Document applied treatments and experimental design layout

Key Data:

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Treatments:
1. Untreated Check (Control)
2. Warm Safflina + Electric Safflina (Test)
   - Rate: 2 L/ha at planting
   - Rate: 1 L/ha at flowering
   - Application: Drip line

Experimental Design:
- Replicates: 6
- Plot Size: 5 ft × 40 ft
- Application: 100 GAL/AC
```

Embedded Content:

- 🖼️ **1 Image** - Field layout map (manually created or auto-generated from coordinates)
- 1 Merged cell range

Process:

- ✋ **40% Manual** - Treatment formulation and application method
- 🤖 **60% Automatic** - Rate calculations, replication matrix, treatment assignments

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4. ASSESSMENT DATA SUMMARY (15% Automatic)

Purpose: Raw field measurement data - the foundation of the entire report

Size: 128 columns × 123 rows (large dataset)

Content Includes:

- Harvest dates and counts
- Fruit size classifications (small, medium, large, unmarked)
- Individual plot measurements
- Post-harvest storage evaluations
- Quality ratings (0-10 scale)
- Tissue sample analysis results

Process:

- ✋ **85% Manual** - Data collection in field and laboratory
  - Manual harvest counts
  - Manual fruit sorting by size
  - Manual weighing
  - Manual quality assessments
  - Tissue sample laboratory analysis
- 🤖 **15% Automatic** - Data entry validation and formatting
  - Data type checking
  - Range validation
  - Automatic time-stamping
  - 22 merged cells for section headers

Data Collection Methods:

1. **Harvest Data** - Manual counting and weighing per plot
2. **Post-Harvest Quality** - 8 fruit samples per harvest held in 60°F storage
3. **Quality Ratings** - 0-10 scale assessment (lower = more marketable)
4. **Plant Vigor** - NDVI and NDRE readings (automated sensors)
5. **Tissue Analysis** - Laboratory submission after flowering

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5. CHARTWORK SHEET (80% Automatic)

Purpose: Intermediate calculations and derived data for statistical analysis

Key Features:

- 📐 **48 Formulas** detected
- Formula Pattern: Cumulative addition (rolling sums)

```
Example: E80 = D80 + E76 (Current + Previous)
         F80 = E80 + F76 (Rolling calculation)
```

Content:

- Phytotoxicity percentages
- Treatment means compilation
- Cumulative calculations
- Statistical intermediate calculations

Process:

- ✋ **20% Manual** - Source data from Assessment Data Summary
- 🤖 **80% Automatic** - Formula-driven calculations
  - SUM functions
  - AVERAGE functions
  - Cumulative totals
  - Cross-sheet references
- 19 Charts automatically generated from this data
- 7 Merged cells for formatting

Automation Method:

- Excel formula references to Assessment Data Summary
- Calculations trigger automatically when source data updates
- Charts linked to formula ranges (auto-update)

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6. FIGURES SHEET (90% Automatic)

Purpose: Visual presentation of all trial data

Content:

- **19 Embedded Charts** showing:
  - Yield by harvest
  - Fruit size distribution
  - Quality ratings over time

    - Treatment comparisons
    - Phytotoxicity progression
    - Statistical significance graphs

Technical Details:

- 163 Merged cells (extensive formatting for chart layout)
- Dimensions: B4:T300 (large chart area)
- All charts linked to Chartwork sheet data

Process:

- ✋ **10% Manual** - Chart selection and formatting preferences
- 🤖 **90% Automatic** - Chart generation and updating
    - Auto-generated when data is updated
    - Formatting applied via template
    - Legend and axis labels automatic
    - Title templates with variable insertion

Chart Types Used:

- Bar charts (treatment comparisons)
- Line charts (trends over time)
- Column charts (categorical data)
- Combination charts (multiple metrics)

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7. POST-HARVEST CHARTWORK SHEET (90% Automatic)

Purpose: Detailed analysis of storage and quality retention

Content:

- Weight loss tracking over storage days
- Quality rating degradation
- Storage duration effects
- Treatment effectiveness for post-harvest quality

Features:

- **2 Charts** (post-harvest specific)
- 8 Merged cells
- Dimensions: B10:N30

Key Finding Highlighted:

> "By Day 4 following the first harvest, untreated check tomatoes were significantly less desirable than Warm Safflina + Electric Safflina-treated fruit"

Process:

- 🤖 **90% Automatic** - Formula calculations from source data
- ✋ **10% Manual** - Interpretation and narrative

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8. AOV MEANS TABLE (95% Automatic)

Purpose: Statistical Analysis of Variance - most critical for trial validity

Size: 249 columns × 136 rows (largest sheet)

Content:

- Treatment means (averages)
- Replication statistics (Rep 1-6 data)
- Variance calculations
- Significance levels
- Standard deviations
- Confidence intervals

Statistical Design: RCB (Randomized Complete Block)

Process:

- 🤖 **95% Automatic** - Generated by statistical analysis software
  - ANOVA calculations
  - Mean separation tests
  - P-value calculations
  - Confidence interval computation
- ✋ **5% Manual** - Software parameter selection

Method:

1. **Data Source:** Assessment Data Summary (raw field data)
2. **Processing:** Statistical software (likely SAS, R, or similar)
3. **Output:** Merged directly into Excel (4,919 merged cells for table structure)
4. **Format:** Professional table with headers and footers

Significance Testing:

- Compares Untreated Check vs Warm Safflina treatment
- Tests multiple harvest metrics
- Tests post-harvest storage effects
- Determines statistical significance (P < 0.05)

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9. WEATHER SHEET (50% Automatic)

Purpose: Environmental context for trial conditions

Data Recorded:

- Date range: June 30 - September 16, 2023
- Precipitation (inches)
- Cumulative precipitation
- High/Low/Mean temperatures (°F)
- Humidity levels (%)
- Dew point

Size: 126 rows of daily weather data (L128 rows including headers)

Process:

- 🤖 **50% Automatic** - Automated weather station data
  - Sensor readings (temperature, humidity, precipitation)
  - Automatic data logging
- ✋ **50% Manual** - Data import and validation
  - Import from weather station or NOAA
  - Manual verification of gaps
  - Unit conversion if needed
  - 1 Merged cell for title

Data Source: Likely NOAA/weather station near trial location (Hyrule Kingdom, Hyrule)

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10. PHOTOS SHEET (100% Manual)

Purpose: Visual documentation of trial conditions and results

Content:

- **26 Embedded Images** showing:

    - Plot setup and layout
    - Plant growth stages
    - Symptom expression
    - Harvest operations
    - Fruit quality comparison
    - Storage conditions
    - Trial closure

Process:

- ✋ **100% Manual** - No automation
    - Manual photography during trial
    - Manual image insertion
    - Manual annotation/captioning
    - Essential for trial audit trail

COMPLETE PROCESS FLOWCHART

```
TRIAL INITIATION
    ↓
[MANUAL] Setup Trial Information
    ↓
[MANUAL] Define Treatments & Apply to Field
    ↓
[MANUAL] Take Photographs of Setup
    ↓
═══════════════════════════════════════════════════════════
    ↓
TRIAL EXECUTION (3+ months)
    ↓
[AUTOMATED] Weather Station Records Environmental Data
    ↓
[MANUAL] Field Data Collection
  ├─ Record harvest dates
  ├─ Count & classify fruit
  ├─ Measure weights
  ├─ Take quality ratings
  ├─ Submit tissue samples to lab
  └─ Take photographs
    ↓
═══════════════════════════════════════════════════════════
    ↓
```

```
TRIAL COMPLETION & DATA PROCESSING
    ↓
[MANUAL] Input Raw Data → Assessment Data Summary Sheet
    ↓
[AUTOMATED] Quality Control Validation
    ↓
[AUTOMATED] Generate Chartwork Calculations
    ├─ Run formulas (48 formulas)
    ├─ Calculate rolling sums
    └─ Compile treatment means
    ↓
[AUTOMATED] Run Statistical Software
    ├─ Input Assessment Data Summary
    ├─ Run ANOVA (RCB design)
    └─ Generate AOV Means Table
    ↓
[AUTOMATED] Auto-Generate Visualizations
    ├─ Create 19 Figures charts
    ├─ Create 2 Post-Harvest charts
    └─ Update all formatting
    ↓
[MANUAL] Interpret Results & Write Narrative
    ├─ Review all statistics
    ├─ Identify significant differences
    ├─ Assess practical significance
    └─ Write conclusions
    ↓
[MANUAL] Final Quality Review
    ├─ Check all calculations
    ├─ Verify data integrity
    ├─ Ensure consistency
    └─ Approve for submission
    ↓
FINAL REPORT DELIVERED
```

AUTOMATION BREAKDOWN

By Component Type:

Component	% Automatic	Details
Data Collection	15%	Field measurements mostly manual; weather automated
Data Entry	20%	Manual input with validation checks

Component	% Automatic	Details
Calculations	85%	Formulas (Chartwork) + Statistical software (ANOVA)
Visualizations	90%	Charts auto-generated from data
Analysis/Interpretation	0%	Manual expert review required
Documentation	50%	Photos manual; metadata semi-automatic
Report Assembly	80%	Template-based with automated linking

Overall Report: ~70% Automatic, 30% Manual

KEY TECHNOLOGIES & TOOLS USED

1. Excel Template System

- Pre-built sheet structure
- Named ranges for calculations
- Cross-sheet formulas
- Conditional formatting rules
- Template locking for data protection

2. Statistical Software

- Input: Assessment Data Summary (raw data)
- Processing: ANOVA with RCB design
- Output: AOV Means Table (automatic export to Excel)
- Likely tools: SAS, R, JMP, or similar

3. Data Validation

- Linked drop-down lists
- Range validation rules
- Data type enforcement
- Conditional cell formatting

4. Chart Engine

- 19 charts in Figures sheet
- 2 charts in Post-Harvest sheet
- Data-driven chart ranges
- Automatic legend generation

- Template formatting applied

5. Embedded Assets

- 1 Field map image (Treatment List & Map)
- 26 Trial photographs (Photos sheet)
- Chart objects (40 total)

CRITICAL MANUAL CHECKPOINTS

Even though 70% is automatic, critical manual quality gates include:

1. **Data Entry Verification** (Week 1 post-harvest)
   - Spot-check field data against raw notes
   - Verify replication counts match
   - Check for outliers
2. **Statistical Review** (Week 2)
   - Confirm design validity (RCB with 6 replicates)
   - Check ANOVA assumptions
   - Review p-values and significance levels
3. **Visual Inspection** (Week 3)
   - Review all 21 charts for accuracy
   - Verify axis scales and data ranges
   - Check legend completeness
4. **Narrative Alignment** (Week 4)
   - Ensure Narrative matches statistical results
   - Verify key findings are statistically supported
   - Cross-reference treatment effects
5. **Final Audit** (Week 5)
   - Study Director signature
   - Compliance with protocol
   - Data integrity certification

TRIAL SUMMARY

Trial Objective: Evaluate "Warm Safflina + Electric Safflina" for yield enhancement in fresh market tomatoes (Tulare variety)

Design:

- **Treatments:** 2 (Untreated Check vs Warm Safflina treatment)
- **Replicates:** 6
- **Design:** RCB (Randomized Complete Block)
- **Duration:** June 30 - September 16, 2023 (2.5 months)
- **Location:** Hyrule Kingdom, Hyrule, California

Key Results:

1. **Yield:** No significant differences between treatments
    - 4 harvests conducted
    - Fruit counts and weights recorded by category
    - Yields similar overall
2. **Plant Vigor:** No significant differences
    - NDVI readings: uniform across treatments
    - NDRE readings: uniform across treatments
    - Phytotoxicity: Minimal (treated plants safe)
3. **Fruit Quality (Post-Harvest):** SIGNIFICANT DIFFERENCE FAVORING TREATMENT
    - Day 4 after harvest: Treated fruit significantly more desirable
    - Weight loss: Marginal and similar between treatments
    - Quality ratings (0-10 scale): Treated fruit rated better
    - Difference maintained through harvests 3 & 4
4. **Tissue Analysis:** Results similar between treatments

CONCLUSIONS

This report is a **modern agricultural template-driven system** where:

✅ **Highly Automated:**

- Spreadsheet calculations (formulas)
- Statistical analysis (ANOVA software)
- Chart generation (Excel charts)
- Table formatting (merged cells + styles)
- Data validation (rules)

⚠️ **Requires Expert Manual Work:**

- Field data collection (no way to automate)

- Result interpretation (requires agricultural knowledge)
- Narrative writing (requires expertise)
- Quality assurance (critical for regulatory compliance)
- Professional judgment (significance vs. statistical significance)

Time Investment:

- Data Collection: 80+ hours (field work)
- Data Entry: 10-15 hours (manual entry + validation)
- Statistical Analysis: 2-3 hours (software runs, review)
- Report Generation: 3-5 hours (chart formatting, narrative)
- Quality Review: 5-10 hours (thorough audit)
- **Total: ~100-125 hours** (mostly in field collection)

Repeatability:

- Once the template is created, subsequent trials follow the same structure
- 70% of the work is repeatable without modification
- 30% is unique per trial (data, interpretation, conclusions)